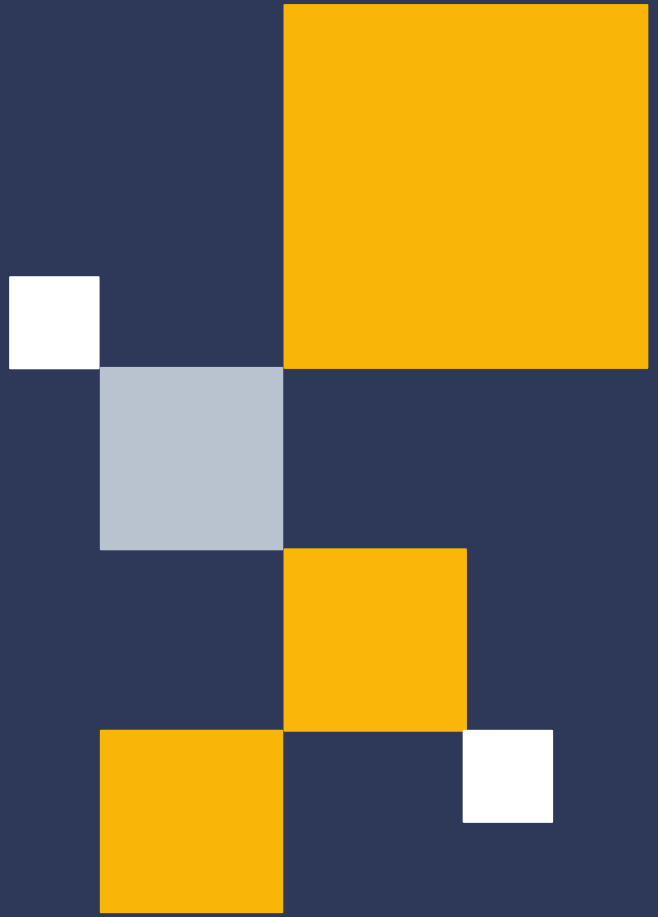
paymenttools

Predict, Don´t Page

AI for Incident Forecasting & Safe Auto-remediation

Rajat Gupta, Engineering Manager



Rajat Gupta



- **Engineering Manager @Paymenttools**
  - Payment Service Provider
  - Fintech-Venture by REWE Group
- **10+ Experience in Engineering**
  - Gaming, Entertainment, E-Commerce, Finance, Consulting
- **Leading SRE and Platform Teams for 4 years**

@rajatguptarg

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@https://medium.com/mr-dops

The Problem

- Reactive pages arrive after damage is done

Insights

- ~350 Alerts per month
- Translates into ~30 incidents
  - Incidents have financial impact
  - Real money is involved

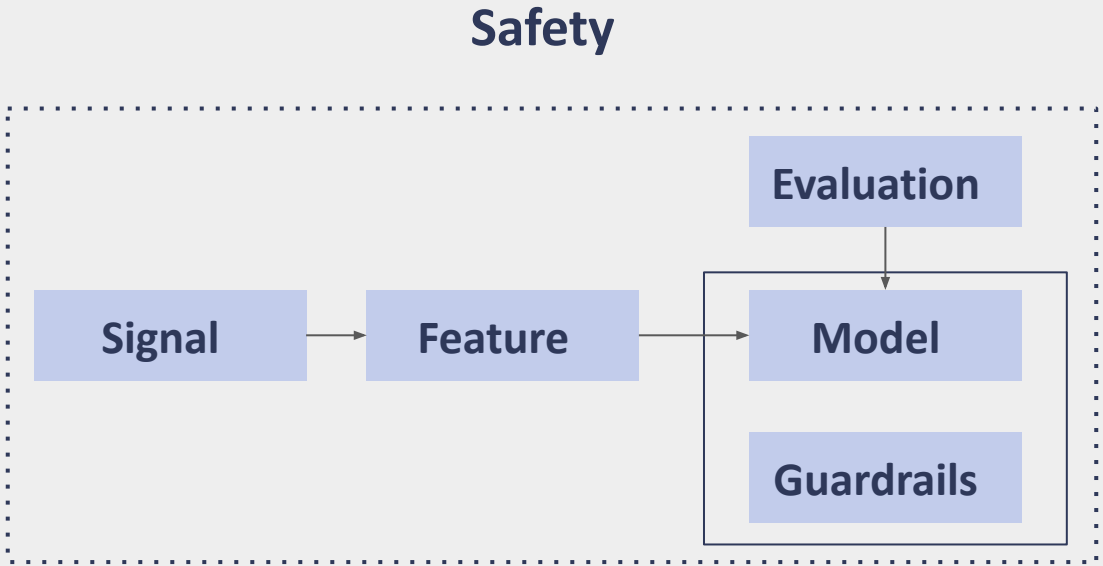


Shift Left

The goal is to get a reliable lead time and act earlier.



Philosophy



Signals

Signal
- Latency
- Errors
- Saturation etc.

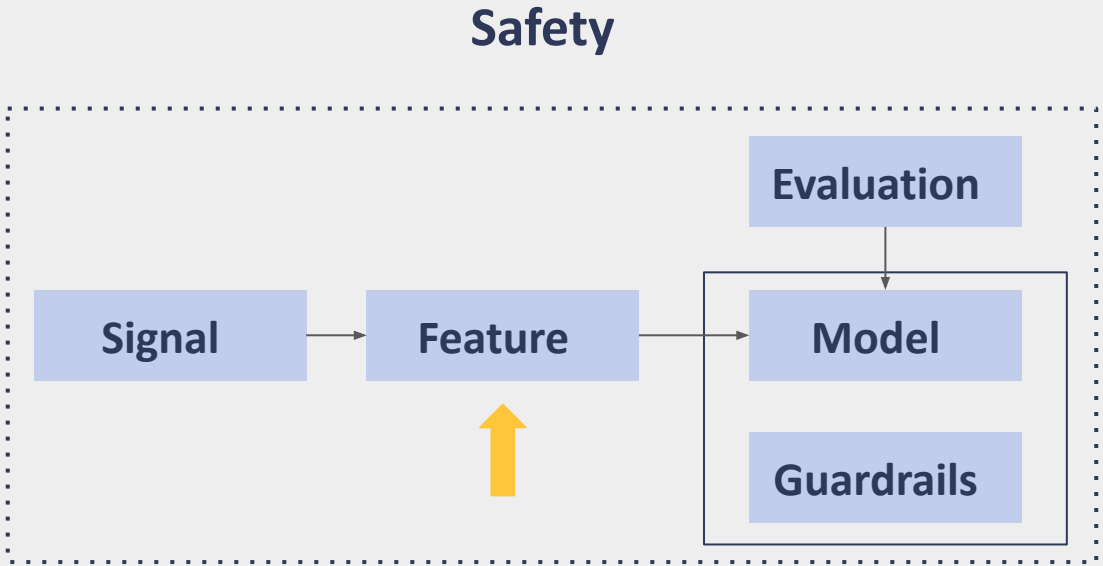
Traces
- Critical paths
- Span errors

Logs
- Errors
- Rate shifts ect.

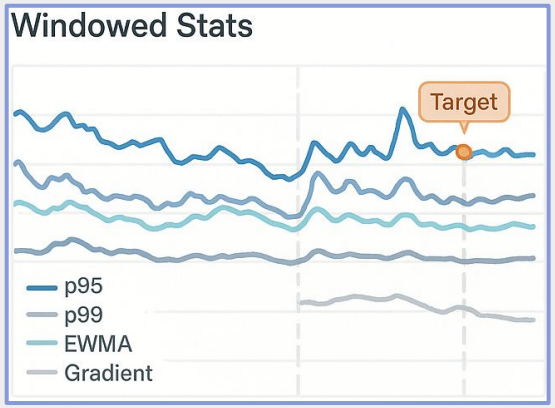
Change
- Deploy diffs
- Feature flags

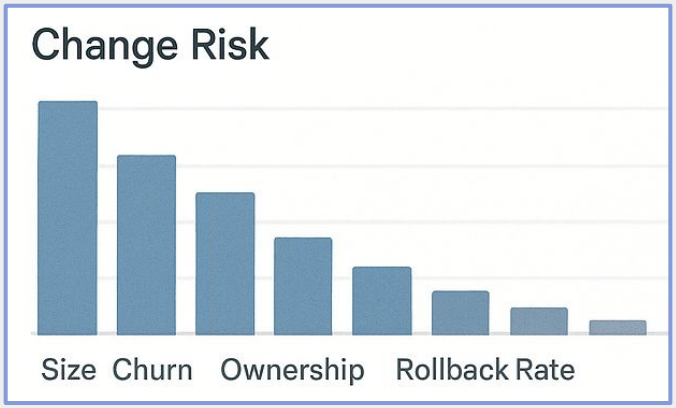
SLO Burn Rate
Budget spent velocity

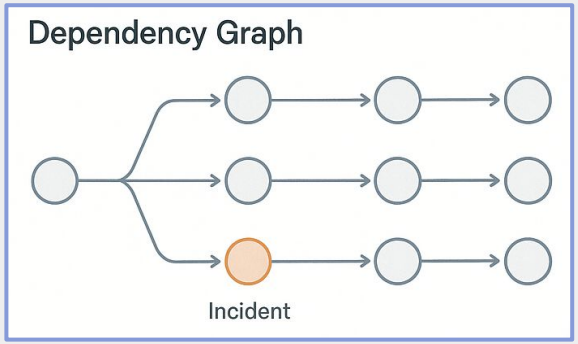
Translating Signal to Feature



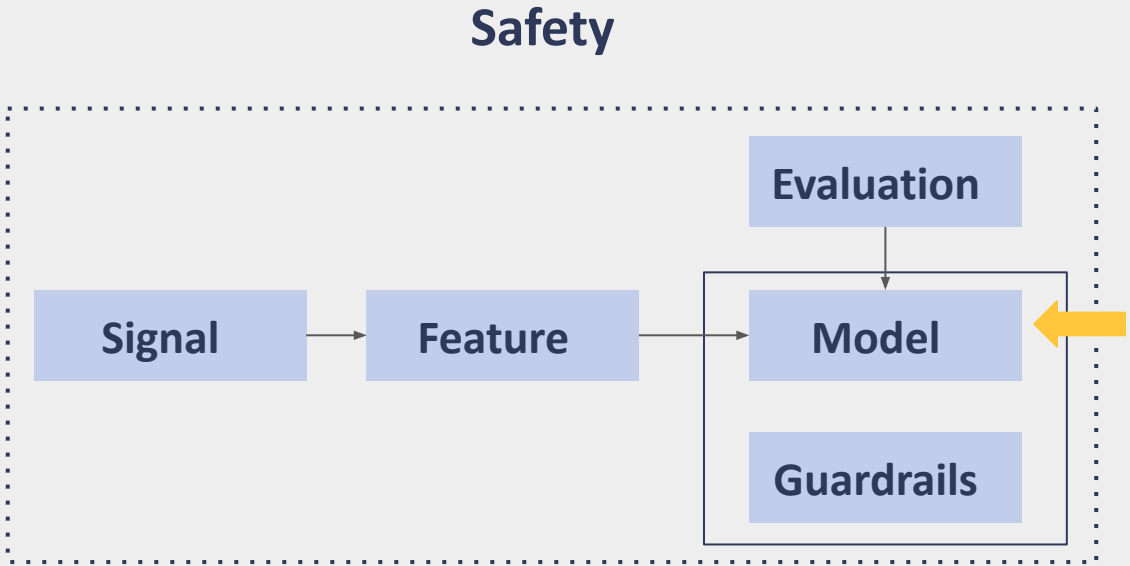
Translating Signal to Feature







Model that actually works



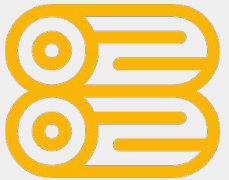
Model that actually works



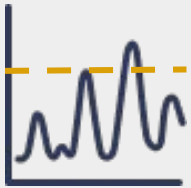
Time Series anomaly detection
for metrics



Logistic regression or gradient trees
for change risks



Log embeddings
for novelty & drift



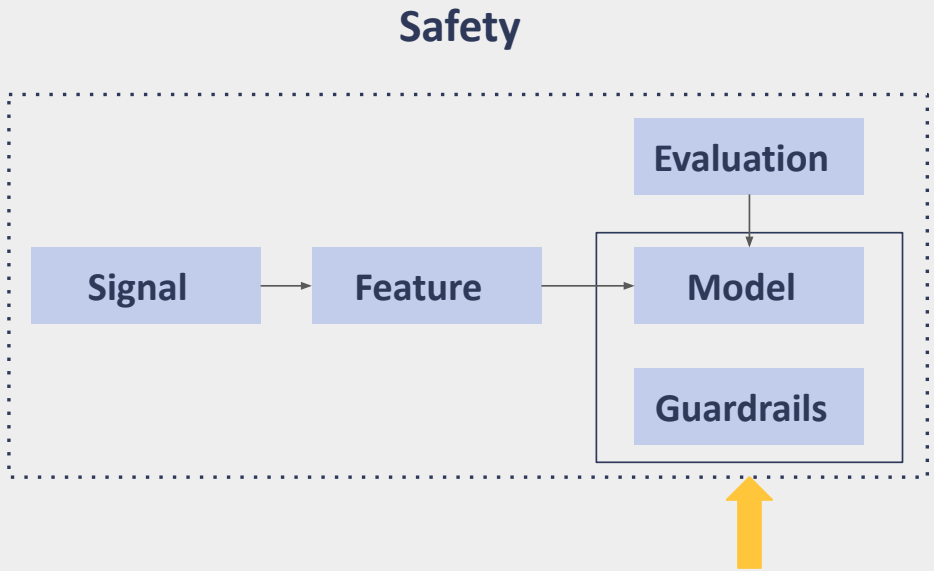




Start Interoperability and Straight Forward

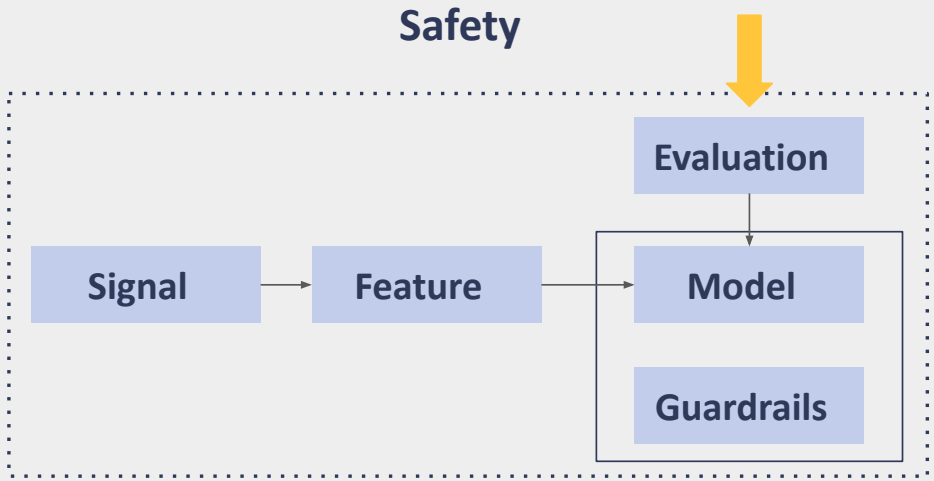
Action Engine Guardrails

- SLO and burn checks
- Blast radius limits by namespace or service
- Approvals or chat ACKs for risky action
- Time-boxed rollback and kill switch



Evaluate the right things

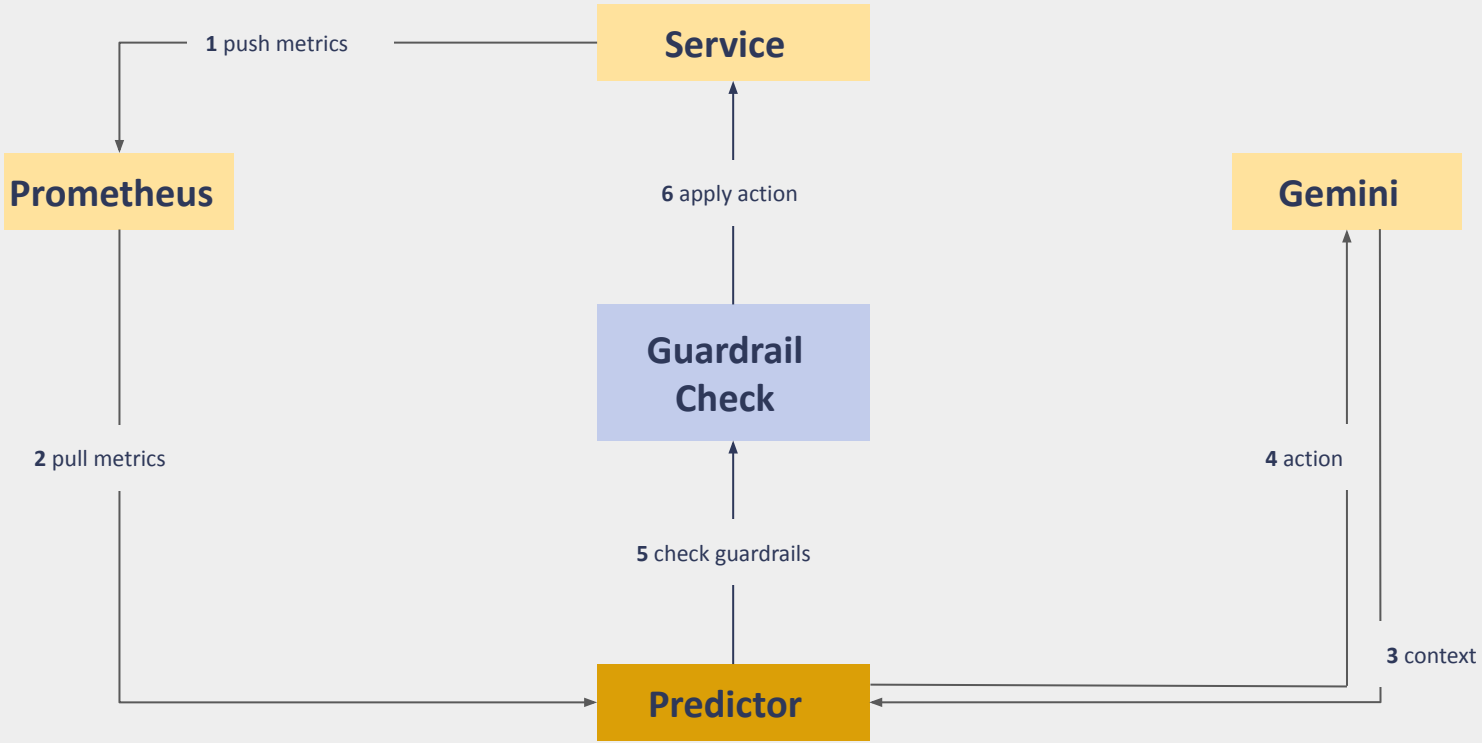
- Precision, recall, F1 and lead time quality
- False positives per week per service
- Cost of action and cost of inaction
- Human review rate and automation trust



Deep Dive



Architecture



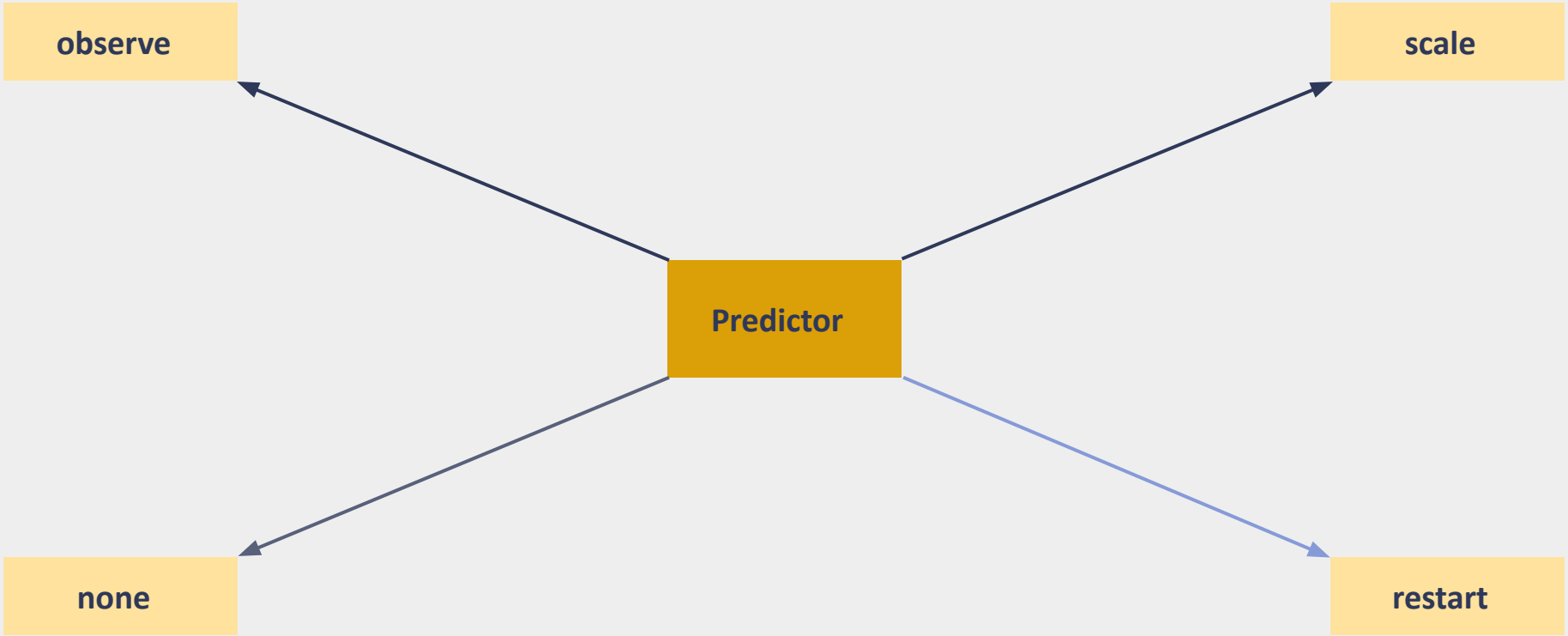
Predictor looks for

- P95 Metric
- Error Rate
- Mean and Standard Deviation
- Z-Score to detect Outlines and Anomaly
- Threshold Values
  - Z Threshold
  - Error Rate Threshold

Context provided to AI

- Current Snapshot of the Signals
- Z-Score Threshold
- Error Rate Threshold
- Recent History of Snapshots
  - Adjustment based on model´s context window and token limits
- Deployment
- Namespace

Architecture



How predictor works

```
[2025-10-16T16:13:28.929879Z] DEBUG: p95 query result: 0.0
[2025-10-16T16:13:35.257124Z] DEBUG: error rate query result: 0.0
[2025-10-16T16:13:35.806294Z] DEBUG: mean/std query result: 0.0/0.0
[2025-10-16T16:13:36.506630Z] {"p95": 0.0, "mean": 0.0, "std": 1e-06, "z": 0.0, "err": 0.0, "ts": "2025-10-16T16:13:36.506583Z"}
[2025-10-16T16:13:37.034973Z] Prediction: normal.
```

Polls the metrics, evaluate p95, error rate, z score and ask the AI with provided context. AI sends back the prediction and recommended action to take.

Detect Errors

```
[2025-10-16T16:14:30.181347Z] DEBUG: p95 query result: 0.9363036201750472

[2025-10-16T16:14:30.506452Z] DEBUG: error rate query result: 0.4445397244700921

[2025-10-16T16:14:30.811239Z] DEBUG: mean/std query result: 0.0/0.0

[2025-10-16T16:14:31.087552Z] {"p95": 0.9363, "mean": 0.0, "std": 1e-06, "z": 936303.62, "err": 0.4445, "ts": "2025-10-16T16:14:31.087386Z"}
```

Get Action to Remediate Errors

[2025-10-16T16:14:30.181347Z] DEBUG: p95 query result: 0.9363036201750472

[2025-10-16T16:14:30.506452Z] DEBUG: error rate query result: 0.4445397244700921

[2025-10-16T16:14:30.811239Z] DEBUG: mean/std query result: 0.0/0.0

[2025-10-16T16:14:31.087552Z] {"p95": 0.9363, "mean": 0.0, "std": 1e-06, "z": 936303.62, "err": 0.4445, "ts": "2025-10-16T16:14:31.087386Z"}



[2025-10-16T16:14:46.524211Z] AI decision: {"enabled": true, "action": "restart", "rationale": "The latest error rate of 0.4445 is significantly above the 0.01 threshold. Additionally, the p95 latency z-score of 936303.62 far exceeds the threshold of 1.0. The sudden spike in both metrics from previous healthy states suggests a critical service failure. A restart is the most direct action to attempt to restore a healthy state."}

[2025-10-16T16:14:46.815535Z] Executing restart remediation

Monitor Recovery

[2025-10-16T16:14:46.524211Z] AI decision: {"enabled": true, "action": "restart", "rationale": "The latest error rate of 0.4445 is significantly above the 0.01 threshold. Additionally, the p95 latency z-score of 936303.62 far exceeds the threshold of 1.0. The sudden spike in both metrics from previous healthy states suggests a critical service failure. A restart is the most direct action to attempt to restore a healthy state."}

[2025-10-16T16:14:46.815535Z] Executing restart remediation



[2025-10-16T16:15:02.164465Z] DEBUG: error rate query result: 0.3603994451778716

[2025-10-16T16:15:17.499763Z] DEBUG: error rate query result: 0.32065217391304346

[2025-10-16T16:15:32.863140Z] DEBUG: error rate query result: 0.041097641830791176

[2025-10-16T16:15:48.184826Z] DEBUG: error rate query result: 0.23627612022595457

[2025-10-16T16:16:03.484081Z] DEBUG: error rate query result: 0.0

[2025-10-16T16:16:03.828874Z] Recovery detected within time box.

Safety by Design



Policy as Code



Staged Rollouts and Shadow Mode first



Deterministic Runbooks and Idempotency



Clear Ownership and Escalation Paths

Ethics & Risks

- Explainability
- Privacy and Retention Limits
- Fail-Safe Defaults
- Manual Override



Outcome

- High Precision, high Recall
- Getting Desired Lead Time
- Less False Positives per week, per service
- Less Human View Rate
- More Automation

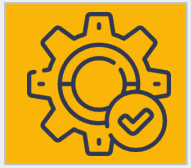
Adaption Playbook



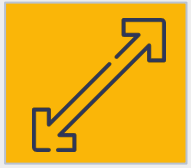
Pick one service and two Use Cases



Shadow Mode with weekly Review



Automate on a safe Runbook with Approvals



Expand the Scope after a green month

Q & A

?



paymenttools

Let´s talk!

rajat.gupta@paymenttools.com